VICTORIA UNIVERSITY OF WELLINGTON TE HERENGA WAKA

AIML231— Techniques in Machine Learning

Week 1

Machine Learning Overview

Dr Qi Chen

School of Engineering and Computer Science

Victoria University of Wellington

Qi.Chen@vuw.ac.nz

Week Overview

★ AI and Machine Learning

★ Machine Learning Scope: Data, Task, Model, and Algorithm

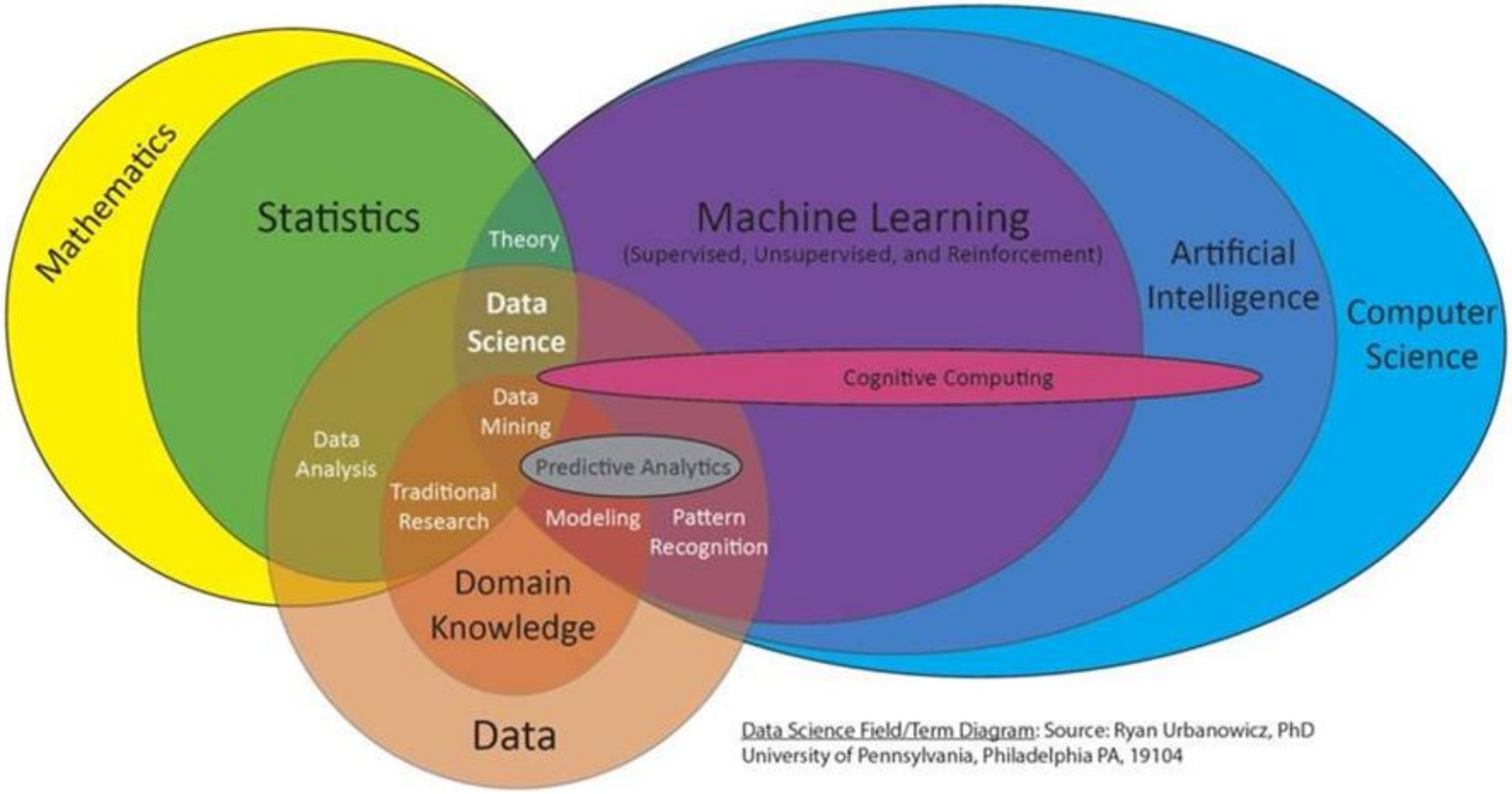
★ Data in Machine Learning

★ Machine Learning Tasks

★ Python Programming for ML Basics

Artificial Intelligence and Machine Learning

- Artificial Intelligence is a broad field of computer science focused on creating systems/machines being able to carry out tasks in a way that we would consider “smart”.
- Machine Learning is a current application of AI based around the idea that giving machines access to data and let them learn for themselves.

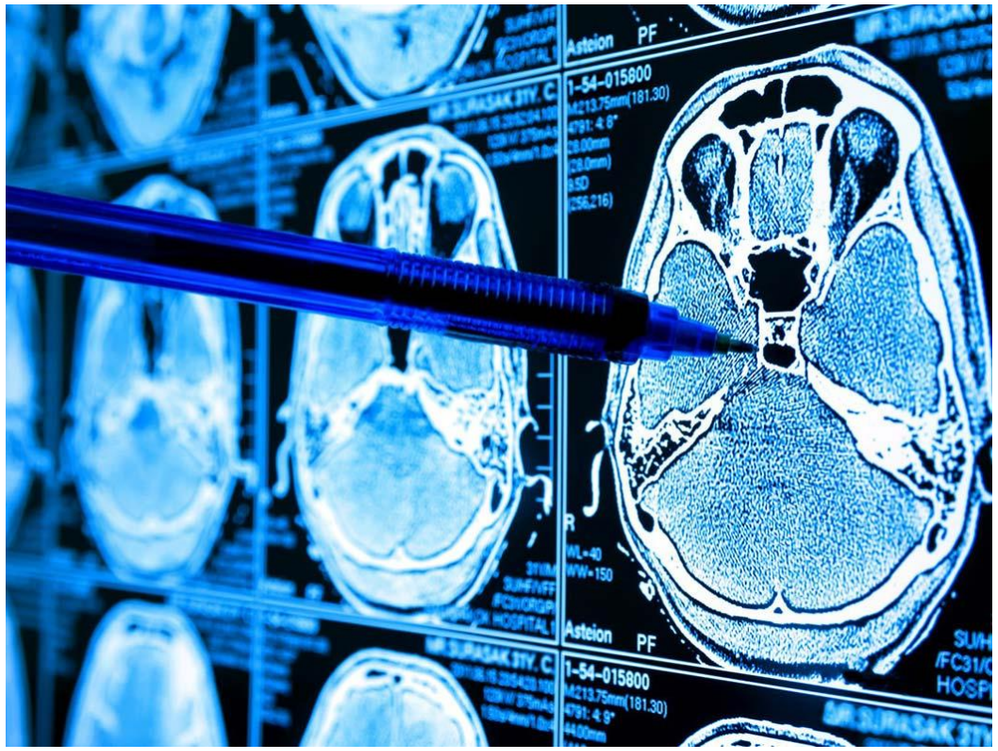


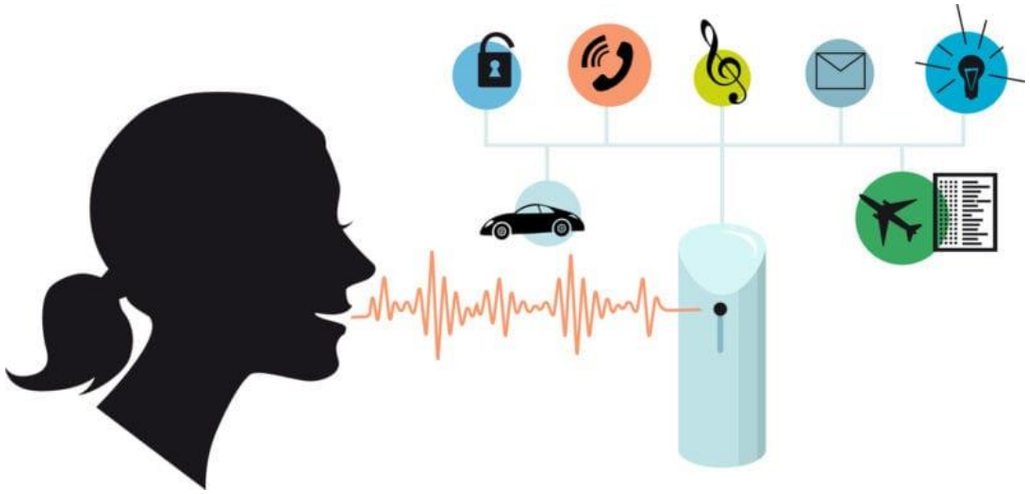
Data Science Field/Term Diagram: Source: Ryan Urbanowicz, PhD University of Pennsylvania, Philadelphia PA, 19104

Machine Learning

- science of getting computers to act without being explicitly programmed.
- a branch of artificial intelligence focuses on the development of algorithms and statistical models
- based on the idea that systems can learn from data, identify patterns and make decisions with minimal human intervention.
- a method of data analysis that automates analytical model building.

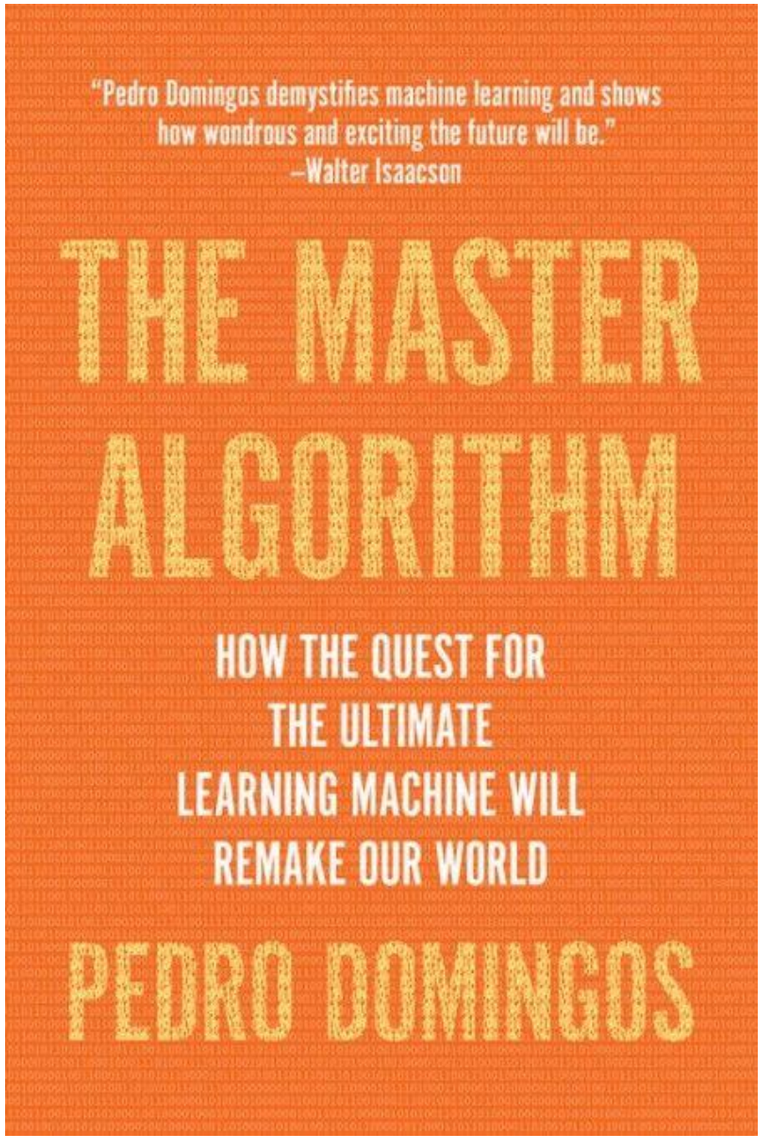
Machine Learning Applications

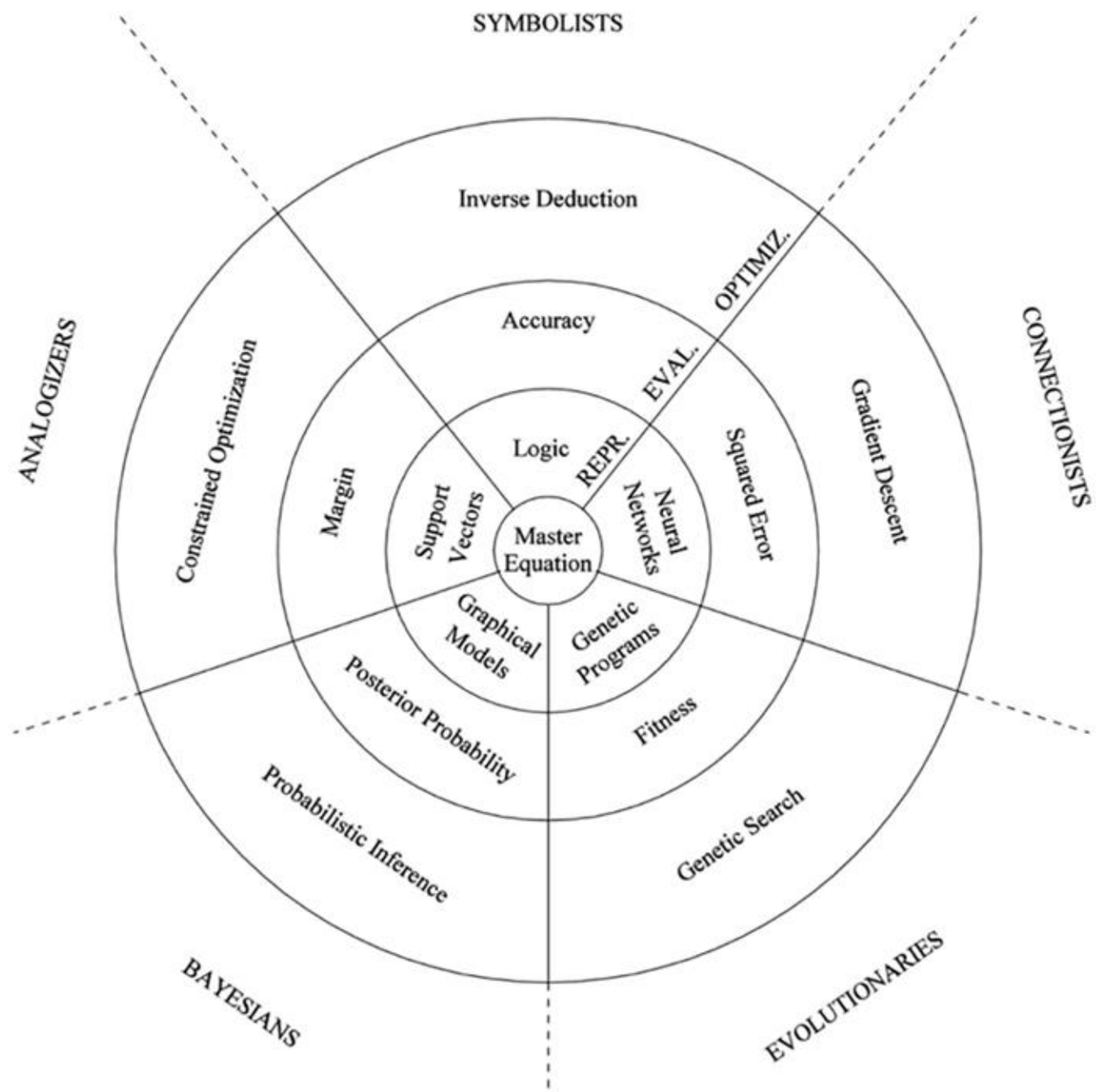




http

Five Tribes of Machine Learning





Domingos, Pedro. "A few useful things to know about machine learning." *Communications of the ACM* 55, no. 10 (2012): 78-87.

The Scope of Machine Learning

ML involves a *wide variety* of each of these:

- data
- task
- model
- algorithm

- Today we'll address a couple of aspects of data.
- Next lecture: the tasks in Machine Learning
- Then in the following weeks: on to the most common models and algorithms

Data

- datahub.io
- openml.org
- Kaggle
- UCI Machine Learning Repository

e.g.

  - iris -- https://www.openml.org/d/61
  - penguins -- https://www.openml.org/d/42585
  - diabetes -- https://www.openml.org/d/37
  - banknotes -- https://www.openml.org/d/1462
  - …

Data

- there are lots of new tools all the time, but near-generic tools at the moment:
  - Python
  - Numpy
  - Pandas
  - Matplotlib
  - Scikit-learn (sklearn)
  - Jupyter notebooks

Data

- Consider banknotes.csv:
- V1-V4 are values of 4 “features”
- the Class is 1 (legit) or 2 (forged)
- Common to talk separately about X and Y:

V1	V2	V3	V4	Class
3.6216	8.6661	-2.8073	-0.44699	1
4.5459	8.1674	-2.4586	-1.4621	1
3.866	-2.6383	1.9242	0.10645	1
3.4566	9.5228	-4.0112	-3.5944	1
0.32924	-4.4552	4.5718	-0.9888	1
4.3684	9.6718	-3.9606	-3.1625	1
3.5912	3.0129	0.72888	0.56421	1
2.0922	-6.81	8.4636	-0.60216	1
3.2032	5.7588	-0.75345	-0.61251	1
1.5356	9.1772	-2.2718	-0.73535	1
1.2247	8.7779	-2.2135	-0.80647	1

X

V1	V2	V3	V4
3.6216	8.6661	-2.8073	-0.44699
4.5459	8.1674	-2.4586	-1.4621
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Y

Class
1
1
1
1
1
1
1
1
1
1
1

~1300 rows in this case. The “Class=2” ones are further down.

Data as "Vectors" in a "Space"

- each row is one data item, here consisting of a pairing:

V1	V2	V3	V4
3.6216	8.6661	-2.8073	-0.44699

Class
1

$$X \rightarrow y$$

- we might talk about **X** = (x_0, x_1, x_2, x_3) instead of the names v_1, etc specific to this dataset.
- ***X*** is a 4-dimensional *vector*
- x_i is the value for the ith *dimension*

Data as points in a space

A “row” can be thought of as a “point” in a “space” of data

It’s easy to visualise when dimensionality is low:

- 1 dimension, e.g. just V1
- 2 dimensions, easy
- 3 dimensions, easy

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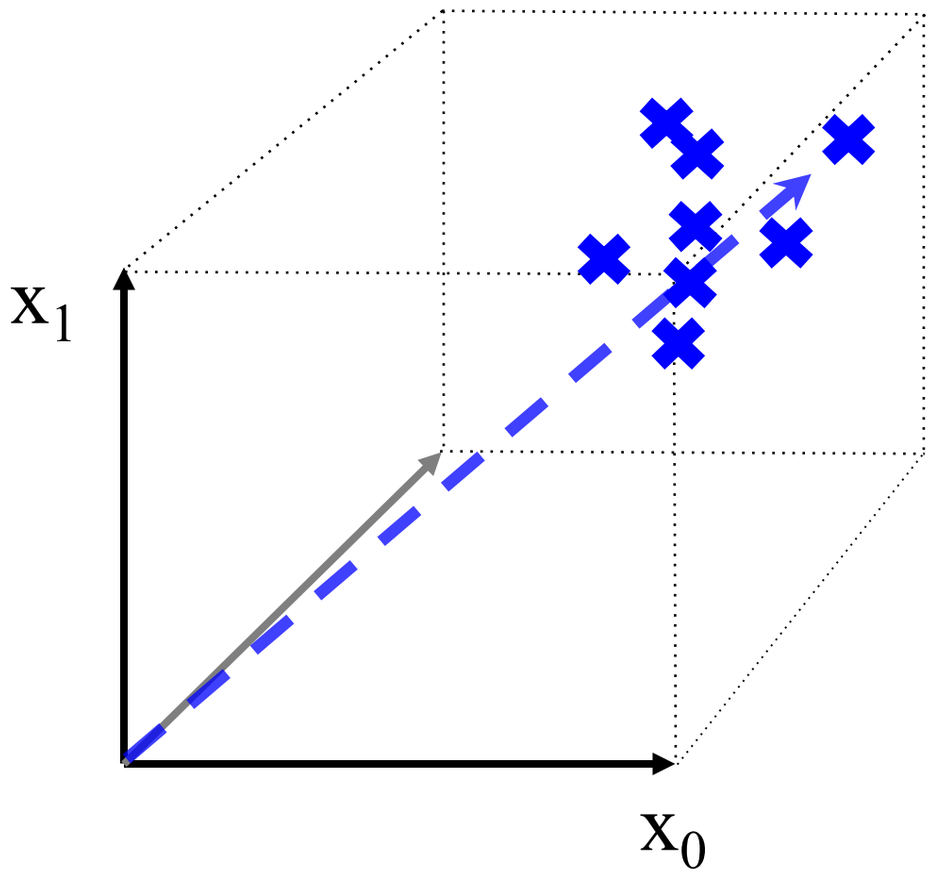
Data as Points in a Vector Space



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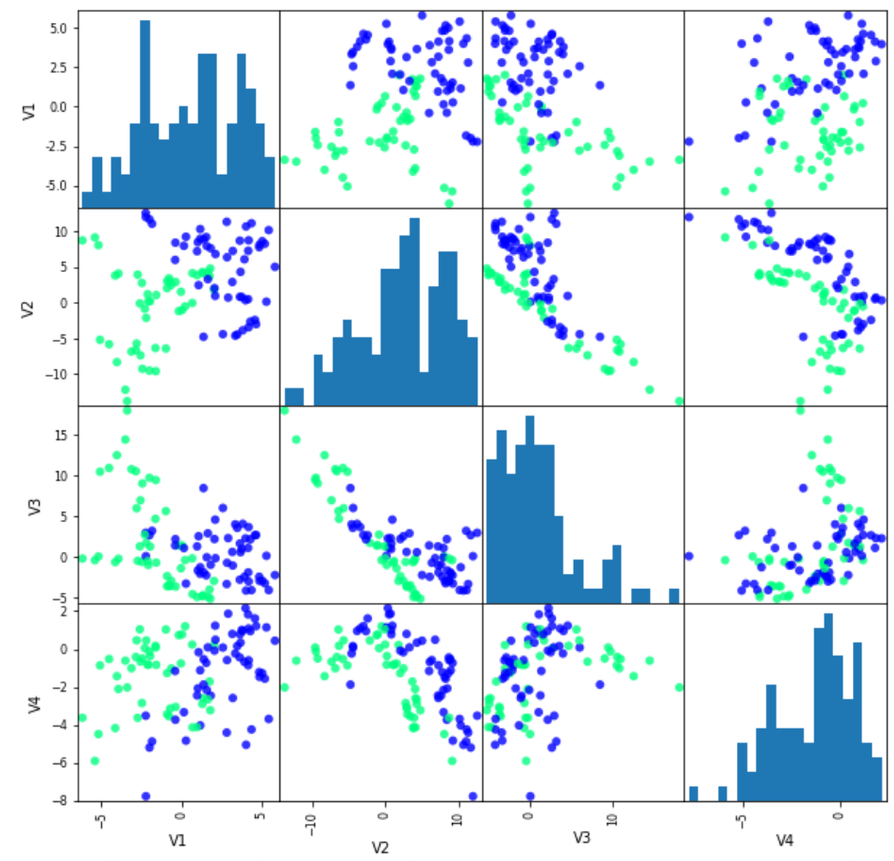


https://sporf.neurodata.io/demos/openml_banknote

Humans can't see in more than 3d

Some of our intuitions hold, some fail

Data in High dimensions

```
for dims in range(1,21):
    print(dims, countInside(1000000,dims))
```

```
1 1000000
2 785776
3 523488
4 307377
5 164873
6 80816
7 37145
8 15879
9 6370
10 2581
11 915
12 324
13 113
14 41
15 11
16 3
17 1
18 0
19 0
20 0
```

none

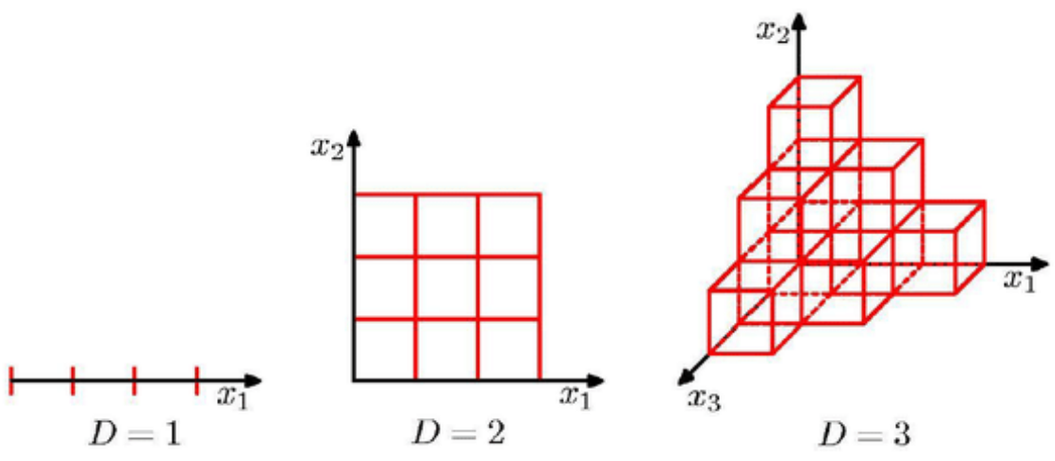
Eg: I made a million data points that were *d*-dimensional, with each dimension being randomly chosen in range -1 to +1. i.e. in a “box” of side 2.

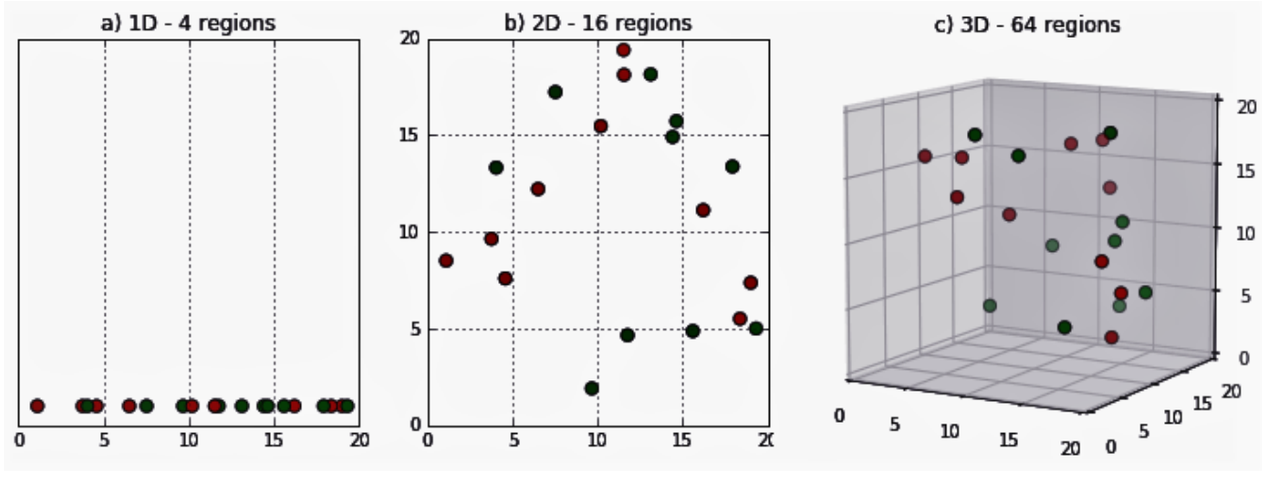
A ball with diameter 2 fits snugly inside this box. Out of a million random points, how many land inside the ball?

The Curse of Dimensionality

Challenges and limitations when dealing with high-dimensional data

- data sparsity, difficult to find meaningful correlations in data
- training the model becomes much slower
- multicollinearity：two or more variables are found to be highly correlated with one another





https://medium.com/@soumikdhar222/curse-of-dimensionality-771dce1d1fbc